The Caterpillar Effect

Final Report

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INTRODUCTION

With over 95,000 employees and $38 billion in revenue, Caterpillar Inc. has expanded from its California roots into an internationally recognized symbol of quality machinery. Caterpillar's major product lines are Construction, Mining Equipment, Diesel & Natural Gas Engines, Industrial Gas Turbines, and Diesel-Electric Locomotives [1]. The departments involved in the project are Global Information Services (GIS) Division and Identity and Access Management (IAM), which aim to eliminate reliance on passwords by 2020.

Over time, passwords have grown in length and complexity in order to meet security requirements. This makes it more difficult to create and remember passwords, leading to exasperated users, and an increased demand on call centers with nearly 4,000 password related help desk calls per month. The use of passwords as the main source of authentication also lead to potential security breaches, which for a company like Caterpillar cost an average amount of $3.6 million [2]. As a result, Caterpillar is exploring the idea of implementing password alternatives for their authentication system.

In a competitive marketplace, user satisfaction is crucial and will lead to success for Caterpillar. One of the most important parts of user adoption is the first time user experience, which needs to be easy to adopt. There are three primary user types accessing Caterpillar's web services systems, which includes both mobile and desktop applications. These groups are

1. Caterpillar customers interfacing with Caterpillar systems;
2. Caterpillar workforce accessing Caterpillar systems; and
3. Caterpillar suppliers and dealers accessing Caterpillar systems.

Caterpillar has over one million users globally, with a large global footprint spanning over 190 countries. These varying geographic locations lead to different demands on which type of authentication system can be implemented, including factors such as noise, dirt, contaminants, weather, connectivity and other regional issues. Due to this, any new authentication system should have the flexibility to accommodate users in these locations, provide several options to the end-user, and allow them to choose the authentication method that best fits their needs.

These authentication solutions must be stronger than passwords, as well as easy to monitor. It is a high priority that the solution in place is compliant with industry standards (i.e. Fast IDentity Online - FIDO[3]) and therefore easier to install and secure into existing identity frameworks by following general industry expectations [4]. This also allows for the technology to be easily replaceable later on if needed, another high priority. Also, it is key to reduce redundancy in the solution, including unnecessary costs and complexities.

OBJECTIVE

As Caterpillar has reached out to QUEST for aid in the initial stage of this authentication overhaul, the team, The Caterpillar Effect, has an opportunity to recommend authentication systems and vendors. The Caterpillar Effect will act as a preliminary research and testing team, culminating with recommendations that will indicate clear future steps for Caterpillar. While these recommendations will be detailed in the Deliverables section, the goal is that they will generally answer the following four questions.

- Which companies are the best candidates for proofs of concepts?
- Which solutions best align to which user segments?
- What is the financial impact?
- What are best practices for rolling out changes to users?

These recommendations will fit into Caterpillar's overall time

line for this replacement project, and allow them to conduct proofs of concepts (POC's), select a product and implement the solution. This will fit into their overall timeline for the authentication overhaul pictured below. This timeline allows the team to better determine the limits of what is relevant to the project, and maintain scope throughout it.

Figure 1: Caterpillar Authentication Timeline (Source: Caterpillar Inc.)

The ultimate measure of success for this project would be for Caterpillar to invest in The Caterpillar Effect's recommendations, confirm through their own future testing, and implement the recommended solutions and vendors. Other measures of success include an improvement in user experience satisfaction metrics, such as password reset metrics and system adoption overall. In the long run, the goal is to increase customer retention, have increased customer sign on, and have increasing positive reviews from all stakeholders. Caterpillar will be able to measure success of these recommendations through metrics such as a reduction in help desk calls, money spent, login time, and breaches.

These recommendations will be based on significant research into both existing and new technologies. This research will allow The Caterpillar Effect to weigh and score different

technologies along with the top providers of those technologies to determine the best possible option for Caterpillar's needs.

Additionally, it is crucial that the recommendations account for differences in the client groups for this project: customers, employees, and suppliers & dealers. These stakeholders all interact with authentication services in different ways and in different environments across the globe. Recommendations must be flexible enough to be broadly applicable to be most useful to Caterpillar.

DELIVERABLES

The team will be providing the client with the following deliverables. These are described in more detail in the following Approach section.

Preliminary Research

- Obtaining Caterpillar-specific data from the client
- Ranking technology via feasibility analysis matrix
- Checklist for companies based on requirements and technologies

This information will allow Caterpillar to preview The Caterpillar Effect's findings and supplement with their own research in the coming year.

Survey Results and Analysis

- Questions to gain insight on different stakeholders
- Analysis and findings from survey

By providing the survey results and analysis, Caterpillar will have more insights into how their stakeholders view their current authentication solutions, as well as how they view new authentication technologies.

Prototype

- Prototype for testing different authentication technologies with Caterpillar users
- Repository for web and mobile application
- Data and results from gathered from the prototype

By providing Caterpillar with the authentication prototype, Caterpillar will be able to continue to collect data on new authentication technologies should they desire. They will also be provided with The Caterpillar Effect's analysis of current data collection.

Recommendations

- Technologies and companies based on findings from research, survey, and test bench
- Analysis of 60 companies and offerings
- Financial Model

- Discussion of impacts

The Caterpillar Effect will provide final recommendations of technologies and companies that would best suit Caterpillar's authentication needs. These final recommendations will also include a financial report, as well as the overall impacts Caterpillar would see from implementing these recommendations.

Final Presentation and Report

- Slide deck used for presentation
- Full report

The Caterpillar Effect will also provide the final report along with the final presentation to the client, to allow them to more easily understand the results of the project, and to be able to continue to share and present The Caterpillar Effect's findings internally.

APPROACH

In order to successfully complete all deliverables and provide meaningful impact to Caterpillar, The Caterpillar Effect devised an approach following the standard decision making model, starting with defining the problem, identifying available alternative solutions, evaluating the identified alternatives, making a decision, followed by evaluation of that decision. After finding and analyzing information from many different sources and viewing the problem from multiple angles, the team was better able to recommend specific technologies and companies.

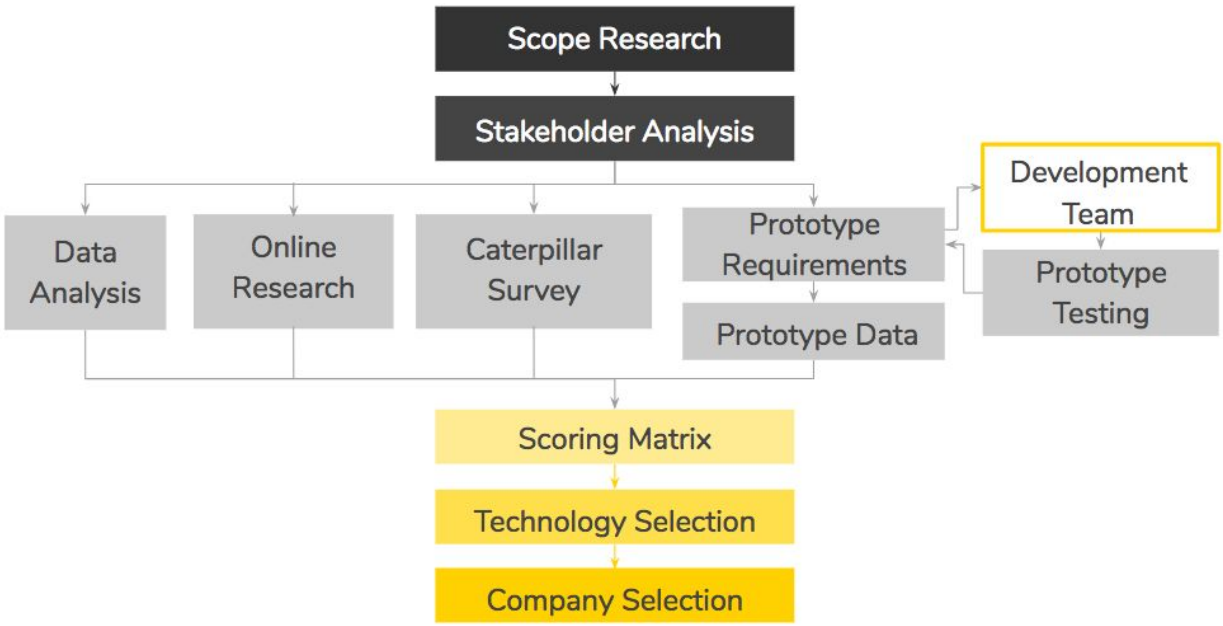

Figure 2. Approach Flow Diagram

Scope Research

The first step in starting this project was understanding the depth of the scope, and determining the opportunity. The Caterpillar Effect utilized resources such as their client, Rowland Nicholson, to fully understand the purpose of the project.

Stakeholder Analysis

Through client calls and meeting stakeholders for the first time through attending the Cat-A-Thon event, The Caterpillar Effect was able to craft a better understanding of the different stakeholder groups within Caterpillar. This included discovering information such as what different stakeholder groups were accessing within the web services accounts, what the roles of each stakeholder were, and different use cases of the authentication web services accounts by each of the stakeholder groups.

Data Analysis

After requesting information from Caterpillar, The Caterpillar Effect looked to understand more about the use of the web services accounts from the different stakeholders. This included information on how often different stakeholders were logging in, where they were logging in from, and who was logging in the most. Analyzing this data allowed the team to further understand how stakeholders are interacting with their current authentication, and other implications authentication has on Caterpillar.

Online Research

In order to create a holistic understanding of the authentication technologies, The Caterpillar Effect conducted extensive online research. The scope of this research included understanding and analyzing all available authentication technologies, global and legal implications of different technologies, industry standards, public perception of technologies, competitor authentication information, companies involved in providing authentication solutions, in addition to background information on methods for evaluating technologies. One of the most valuable sources cited during background research was the Gartner Report [5], which provided information on how to evaluate technologies, and a wealth of other information on authentication technologies.

Through this research, The Caterpillar Effect identified 14 different authentication technologies, including password, text push notifications, facial recognition, fingerprint, one time password, iris scan, heart rate, smart card/RFID, USB key, QR code/ image scanner, behavioral analytics, pattern recognition, voice recognition, and social login [6] - [15].

Feasibility Analysis Matrix

In order to compare each technology, The Caterpillar Effect created a feasibility analysis matrix based off of three main categories- user experience, installability, and security. Within each category, the team created another feasibility analysis matrix in order to assess each category individually. In order to score the various technologies, The

Caterpillar Effect did extensive online research, consulting reputable sources including Gartner's "Best Practices for Selecting New User Authentication Methods" report [16], IBM's "2017 Cost of Data Breach Study" report [2], and PwC's "Moving forward with cybersecurity and privacy" report [17].

For user experience, the technologies were scored on the learning curve, process flow interruption, success rate, time taken for login, number of clicks, and forget-ability. For installability, the technologies were scored on general compatibility with different types of devices, and how easy it would be to integrate with Caterpillar's current authentication system. For security, the technologies were scored on the ability of phishing, password guessing, re-used password attacks, shoulder surfing, theft of authenticator, and channel vulnerabilities. Cost was also another key factor considered in evaluating different technologies; however, due to a lack of pricing estimates and the complex nature of how authentication systems are priced, this information was not able to be integrated as a part of the matrix.

Additional Requirements

Through discussions with our client and other research, the team was able to identify requirements for both technology and company solution. These are as follows: FIDO compliant, multi-factor authentication, single sign-on, Bring Your Own Device (BYOD) policy compliant, cloud-based, scalable, and has analytics reporting.

These additional requirements helped ensure that any recommendations would be of the highest quality and standard.

Competitor Findings

The team also investigated the Identity & Access Management (IAM) and Authentication solutions that various competitors to Caterpillar utilize, including John Deere, Komatsu and Kubota. Additionally, The Caterpillar Effect studied various companies that Caterpillar uses as benchmarks for their IT solutions, including other companies in the Fortune 50, such as Boeing. This allowed the team to supplement their own research by finding real life examples of comparable companies to Caterpillar, who have successfully implemented stronger authentication systems and could provide proof of the viability of these technologies.

Twitter Data Mining

To better understand general public sentiment of different authentication technologies, The Caterpillar Effect took to Twitter to analyze tweets based on certain keywords. The team utilized the TwitterScraper python library to initially pull up to 15,000 tweets from a two week period based on the keywords associated with a certain technology. This generated text files that were then imported into excel to be further analyzed. This

analysis consisted of counts of various keywords that were commonly used in the positive and negative responses from Caterpillar employees to the survey. This data mining allowed The Caterpillar Effect to supplement their secondary research and Caterpillar viewpoint, with a wider range of data from the general public.

Caterpillar Survey

In order to understand the Caterpillar stakeholder's point of view on authentication, the team went right to the source with a survey. By conducting a survey, The Caterpillar Effect was better able to gauge numerous factors to help ensure that any authentication solution recommendation would best satisfy the desires and take into account the input of the different stakeholder groups.

The survey went through multiple rounds of iterations, and was reviewed and approved by both the team's faculty advisor as well as their project champion. The team conducted a preliminary round of surveying Caterpillar stakeholders at a local Cat-A-Thon event on October 10th; this was critical to understand factors to include in the authentication prototype, and how to simplify and improve the survey. The final survey was distributed to Caterpillar stakeholders with the help of Rowland Nicholson. The survey was first distributed on October 25th to GIS newsletter, followed by distribution to the Corporate Yammer community (enterprise social media), and other CAT dealer contacts. In order to help incentivise participation in the survey, The Caterpillar Effect offered participants the ability to enter their email address for the chance to win a $25 Amazon gift card.

Authentication Prototype

The next step of The Caterpillar Effect's approach was testing out different authentication technologies. The team developed prototype requirements and then paired with a computer science capstone team to create the prototype.

The computer science development team consists of Natnael Asfaw, Aerin Cox, Brian Levine, Ian Longo, Dylan Mink, Nick Myers, Michael Weglein, and Jeffrey Yu. The development team was advised by Dr. Purtilo in his CMSC435 class. They typically met with the Caterpillar Effect on Wednesday evenings to design, develop, and update on progress. The prototype had a web/desktop version as well as a mobile app. The final prototype included a range of different technologies, including facial recognition, USB key, fingerprint, one-time password, and regular passwords. The dev team configured the flow of the prototype so that it prompted users to fill out identifiable information, before randomizing three technologies that user would test out. After completing these authentications, the user would then be prompted with a brief survey asking about ease of use, user experience, and if they would recommend the authentication methods they used.

The Caterpillar Effect was able to test the prototype as it was being built in order to ask questions, communicate with the client, and provide feedback in an iterative manner. The

development team was able to call and demo to Rowland Nicholson and other Caterpillar employees during one meeting. The Caterpillar Effect collaborated with Rowland Nicholson in order to send the prototype out to stakeholders and conduct testing. In order to help incentivise participation in the authentication prototype, The Caterpillar Effect offered participants the ability to enter their email address for the chance to win a $25 Amazon gift card.

The development team created a database to collect and store data, such as the user survey questions along with number of clicks and time taken for each user to finish the process. At the end of the testing period, The Caterpillar Effect was able to download the user data to analyze and provide visualization through graphs.

Scoring Matrix + Technology Selection

The team created a hypothesis in which they list ed companies thought to be best suited for Caterpillar needs. Then, to consolidate the findings, the team created a comprehensive scoring matrix, in order to select technologies best suited for Caterpillar. The matrix took a weighted average of the scores from the feasibility analysis matrix, the survey, and prototype. The Caterpillar Effect was able to confirm the hypothesis after doing an holistic analysis of all the data collected using the matrix. By utilizing results from the scoring matrix, The Caterpillar Effect selected a range of technologies that would best satisfy all stakeholder needs.

As described, there was a large variety of background information collected by The Caterpillar Effect in order to be able to recommend the best solutions for Caterpillar. The team evaluated all of these findings through a qualitative analysis, to pick what they felt through a series of discussions and consideration what they felt would be the best authentication technologies. However, simply picking technologies would not be enough to ensure the best solution was truly found. Therefore, the team used a quantitative method of analysis, creating a matrix through a series of scorings, to evaluate on a more technical level all of the different technologies. By proving that recommendations would be the same through the initial qualitative analysis and then through a quantitative analysis, the team was assured that they had developed the strongest possible recommendation for Caterpillar.

Company Selection

Through comprehensive online research, The Caterpillar Effect was able to develop a list of the top 60 companies in the industry that would be able to provide for Caterpillar's authentication needs [18]-[56]. For each company, the team identified which technologies it offered and checked to see if the company offered single sign-on, multi-factor authentication, analytics reporting, compatibility with Cordova, scalability, included in the FIDO alliance, on Cloud. The Caterpillar Effect created a shortlist of companies for Caterpillar to further pursue a relationship with. In addition, the team also provided a second list of companies that would not be the most

compatible with Caterpillar's current needs, but that Caterpillar should keep on its horizon in the future.

Determining Impact

After completing objectives, The Caterpillar Effect used data from research, financial information provided by Caterpillar, and data collected in the authentication prototype to develop a financial model. This model discovered savings that would be seen by Caterpillar if they were to switch to the recommended technologies. In addition, the team detailed other non-monetary impacts. The impact also took into account the timeline for implementation of the project by Caterpillar.

RESULTS

Online Research of Authentication Types

One of the key results of The Caterpillar Effect's online research was to better understand what alternatives exist for authentication. Table 1 best summarizes the advantages and disadvantages determined for each of the authentication technologies.

Table 1: Advantages and Disadvantages of Authentication Technologies

	Advantages	Disadvantages
Password	-Low learning curve -Low implementation cost	-Length & hard to remember -Insecure -High Maintenance costs -Bad user experience
Text Push	-No extra software required -Acts as second factor	-Slow -Inconvenient -Require separate device
Facial Recognition	-Possibility of spoofing -No user input required	-Biometric data security -Camera required
Fingerprint	-Very secure, unique. -Technology widely available	-Requires exposed fingerprint -Difficulties with damaged fingers. -Additional hardware required
Heart Rate	-Unique to each user -Passive authentication	-Developing Technology -Additional hardware Required

	-Potentially changes over time	
Iris Scan	-Unique to each user -Doesn't change over time	-Biometric data security concerns. -Additional hardware required
Voice	-Fast authentication -Hands free	-Easily faked -Difficult to authenticate in loud environments
Behavioral Analytics	-Passive authentication for user -Uses existing devices	-New technology, potentially insecure or immature -Requires training database of users habits
One Time Password	-Second factor of authentication -Very secure	-Possibility of shoulder-surfing -Requires separate device -Requires special software
Pattern	-Easy for user -No additional hardware	-Not as secure as other methods. -Limited to touch screen devices.
Social	-Easier to remember and login.	-Not as secure -User needs account on social network. -New attack vector from social networks.
Smart Card/ RFID	-Very secure physical card or device -Quick authentication	-Possibility of user losing physical card or device. -Additional hardware required.
QR Code/ Image Scanner	-Mature technology -Compatible with wide range of devices.	-Requires multiple devices. -Requires camera hardware.

Feasibility Analysis Matrix

As mentioned in the approach section, the team analyzed various technology solutions on the basis of user experience, installability, and security. As seen in Appendix A, The Caterpillar Effect ranked each solution to find leaders in each category. The results are as follows.

Leaders in Security

- Heart rate
- USB key
- Voice

Leaders in User Experience

- Fingerprint
- Heart Rate
- Behavioral analytics
- Social
- Face
- Iris

Leaders in Compatibility and Installability

- Password
- Text push notifications
- Social Passwords

Overall Technology Leaders

- Fingerprint
- Heart Rate
- USB Key
- Social Passwords

Twitter Data Mining

Pulling Twitter data over a discrete period yielded various ratios of keywords. The team selected those that they believed would have the most clear results, and would minimize incidental usage of the technology keyword in an unrelated message. The ratio the team ultimately used was "hate" to "love". Appendix B includes a chart of the specific numbers we were able to get for the different keywords. However, for only this love/hate ratio, The Caterpillar Effect found that "passwords" resulted in 65.2% hate to love, whereas "fingerprint" was at 31.4%, "facial recognition" at 28.6% and "yubikey" at just 12.5%, emphasizing how the general public feels much less favorably towards passwords than other authentication technologies.

Caterpillar Survey

The initial survey was conducted at the CAT-A-Thon event at the Alban CAT dealership in Baltimore. It was administered as more of a survey/discussion combination. The Caterpillar Effect was able to collect eight in-person survey responses with insight into Caterpillar Inc. users' thoughts on authentication. Moreover, the survey was a little lengthier than the respondents expected. Learning this was very important to the team, as they did not want to potentially frustrate respondents who end up rushing or skipping written answers to finish

quickly. The team evaluated the purpose and intent of their questions and were able to cut down and combine some questions.

The Caterpillar workforce survey closed with 222 respondents. The final survey questions include the following:

- Occupation title
- Country of residence
- Length of employment with Caterpillar
- Sensitivity level of data accessed
- Satisfaction with password security
- Satisfaction with password experience (and why/why not)
- Location of login
- Type of device
- Reason for login
- Number of logins per day
- Tendency to adopt technology
- Technology: previously used and comfortable, previously used and don't like, interested in using, never heard of
- Account compromise
- Suggestions for improving authentication

Some notable statistics include:

- Over 60% of respondents enter their password 6 or more times per day.
- 66% of respondents have been at Caterpillar for 10+ years.
- 87% of respondents have access to yellow or red (confidentiality) information.
- Around 90% of logins were on desktop, 10% on mobile.
- Repeated comments include: too many different passwords to remember, have to reset passwords frequently, security and compromise concerns
- About 90% of respondents tend to adopt new technology fairly quickly, falling into the Innovators, Early Adopters, and Early Majority range, which is ahead of the innovation adoption curve which is typically only 50% (see Appendix F).

Technologies Caterpillar is most interested in:

- Face
- Iris
- Smart Card
- USB Key
- QR Code
- Fingerprint

For more detailed information on the results of the Caterpillar Stakeholder survey, please see Appendix F.

Authentication Prototype

The Caterpillar Effect was able to collect several data points from the prototype including perceptions on user experience, ease of use, and recommendations, in addition to statistics such as time spent on each authentication method and the number of clicks each authentication method took. (Please see Appendices D and E for the full authentication prototype)

The chart below graphs the 221 total responses received from the survey about the authentication user experience. USB Keys and Fingerprint readers received the highest user experience scores, followed by facial, one-time password, and (lastly) password.

Figure 3: Authentication Prototype User Experience Results (Source: The Caterpillar Effect)

Overall, the USB Key (tested as the Yubikey NEO), was a leader in all three categories, proving that Caterpillar employees can quickly master a new technology and were favorable to change. Fingerprint and facial recognition were also favorable in all categories. As expected, password, matching the same requirements instilled by Caterpillar Web Services, scored the lowest in user experience and was the least recommended.

When averaging the scores from all prototype metrics, the results were as follows:

1. USB Key
2. Finger
3. Face
4. One Time Password
5. Password

Scoring Matrix

As seen in Appendix C, the Caterpillar Effect took an average of all scores from the feasibility matrix, survey, and prototype. With the three methods of scoring and performing a weighted average, the results were as follows:

1. Face
2. Social
3. USB Key
4. Fingerprint

The scores for these top 4 technologies ranged from 3 to 3.6. This was the top "cluster," with the other technologies following reasonably farther starting at 4.5.

The team also compiled a watchlist of technologies that are not mature enough yet, but could become very compatible in the future. These technologies include heart rate, behavioral analytics, and bluetooth. Currently, these technologies do not have a proven track record and general social adoption. For advanced biometrics like heart rate technology, it makes sense for Caterpillar to first use basic biometrics first. On the adoption curve, appropriate technologies should not be so early in the scale as to mitigate instead of attract use. Some technologies also have some specific system requirements. Heart rate technology requires user comfort with wearables, and smart cards require security infrastructure to allow technology reliance. There are more appropriate options for Caterpillar's current authentication needs.

Company Matrix

After analyzing over 60 companies, the team recommends that Caterpillar further researching Okta, Bitium, Deepnet, Gemalto, and Secure Auth. Each of these companies offer excellent user experience, with a reputable customer base and partnerships within the industry. They appear to be on the frontlines of innovation and will be able to adapt as authentication technology continues to evolve in the future. More importantly, they meet Caterpillar's vendor requirements, such as being compliant with FIDO standards, offering single sign-on, and being cloud-based.

To see the full analysis of 60 companies, please reference Appendix G. The top five companies that stood out as most compatible with Caterpillar are below:

1. Okta was named a leader in identity management by Gartner. They offer a wide range of authentication alternatives ranging from biometrics, to social, to USB key. In addition to satisfying the vendor requirements above, they are also compatible with Cordova, a framework that allowing for cross-platform code development. In addition, the cost was reasonably low at $2-$3 per user per month [20].

2. Bitium is a provider of enterprise-class single sign-on, identity access management, and password management for cloud-forward companies and teams. They offer various

biometrics, one time passwords, and USB keys. In addition, they match almost all criteria. They offer their services for $8 per user per month [22].

3. Deepnet is a leading vendor in multi factor authentication. They offer biometrics, one time password, smart card, and USB key, while also matching most criteria. They also have a reasonable price at around $36 per user per year ($2.66 per user per month) [29].

4. Gemalto's SafeNet identity and data protection solutions are trusted to protect data, identities, and intellectual property. They offer various technologies such as one time password, smart card, USB Key, and offer bluetooth, which is a technology The Caterpillar Effect hopes to see grow. They match almost all criteria and are most economically feasible with $1 per user per month [45].

5. Secure Auth provide multi factor and single sign-on solutions with adaptive authentication and a frictionless user experience. They provide centerprise-grade security for all levels in the organization. They offer technologies such as one time password, USB Key, and social passwords, but do not offer any biometrics. They do match most criteria and cost about $5 per user per month [18].

In addition to these excellent companies, The Caterpillar Effect compiled a watchlist of companies that the team believes could be successful and compatible in the future. These include Nymi, Nuance, Safran, and SAASPASS [32], [39], [44], [49]. These companies are rapidly growing and investing in watchlist technologies emerging in the market.

Financial Model

It is important to understand an estimation of the financial implications of The Caterpillar Effect's proposal. To measure the project's value, the model hinges on three major factors: workforce efficiency, help desk calls reduced, and implementation costs. For the financial model, workforce efficiency is a measure of time productivity. This is a 'soft' dollar - meaning that it isn't seen on a balance sheet, but instead provides unrealized benefits. To do so, the prototype provided average login speeds of recommended technologies against password. For example, the average login time for passwords is 26.70 seconds and 5.56 seconds for Yubikey. The translates to time saved per login. Looking at the survey data, the average stakeholder logs in more than 6 times a day, equating the time saved per day. In addition, Caterpillar Inc. provided the average salary as $74,000. Converting seconds to minutes and days to years and scaling the difference with the salary, the total is money saved in salary per year per person.

Unlike the workforce efficiency, Caterpillar Inc. can directly relate the number of help desk calls to a balance sheet value. With new technologies expected to reduce help desk calls, Caterpillar Inc. can expect cost savings. The model is based on the following assumption:

Table 2: Projected Help Desk Adoption Rate

		Year			
		1	2	3	4
	% help desk calls increased	15%	30%	40%	50%
No Change	% total calls increased	112%	123%	131%	139%
	% help desk calls decreased	0%	20%	50%	70%
Savings	% total calls reduced	100%	84%	61%	45%

The model assumes that password reset costs will continue to rise with increased password requirements and will decrease with a shift away from passwords. The second assumption is in line with Caterpillar Inc.'s adoption curve as per the stakeholder survey. Looking at Forrester Research estimate, the average help desk call is between $25-40 [58]. After speaking with Caterpillar, representatives; however, the average cost is over Caterpillar's cost. As a result, the model assumes a $20 average help desk call. In addition, there are 44,040 help desk call per user / year. Multiplying the assumed percentage rise or decline with the total help desk calls per year and the cost of the call, the result equates to a reduced cost per year.

The last stage is the implementation costs. These include HR training, software development, technology solution cost, and support costs. HR training costs to communicate the password policy change is roughly $750,000, after discussion with Caterpillar Inc. project managers. The technology solution cost varies on the company; because Caterpillar Inc. is keen on cutting down the sale price, the model assumed that it could achieve $24 per user per person, or $2.4 million scaled to 100,000 users. Rowland Nicholson estimated that development will require 160 hours of salary at a $125 developer burden rate, which is a total of $20,000 upfront cost. Support costs are also generally 15% of the technology solution cost. Combining these upfront and recurring costs with the workforce efficiency and help desk calls reduced. For the three years, the model can discount with an industry standard of 8.06% cost of capital. After year four, the model can assume that there is a terminal savings of 70% of help desk calls reduced, so discount with the year three rate.Summating each cash flow becomes the projects assumed net present value. With The Caterpillar Effect's assumptions, this was roughly $2.2 million, as seen in Table 3.

Table 3: Financial Model

$20 Per Help Desk Call	1	2	3	4
Current Total Cost per Year	$983,853.60	$1,086,907.20	$1,155,609.60	$1,224,312.00
Reduced Cost per year	$880,800.00	$743,395.20	$537,288.00	$399,883.20
(+) Workforce Efficiency	$3,048,777.78	$3,048,777.78	$3,048,777.78	$3,048,777.78
(-) Training	$750,000			
(-) Software Development	$20,000			
(-)Solution Cost	$2,400,000	$2,400,000	$2,400,000	$2,400,000
(-) Support Costs	$360,000	$360,000	$360,000	$360,000
Savings	-$378,169	$632,290	$907,099	$1,113,207
Discounted	-$378,168.62	$585,128.43	$776,828.13	
3 year NPV				
Sum: y1-y3 FCF	$983,787.93			
Discounted Terminal Value	$1,202,931			
NPV Project	**$2,186,718.96**			

RECOMMENDATIONS

After evaluating the different technologies based on the data the team had collected, the final recommendations are represented graphically in the chart below. For all stakeholder groups, fingerprint and facial recognition authentication are recommended. These biometrics are recommended for all stakeholder groups because of their versatility over different platforms, user friendliness and extremely high security. For employees, who most often have access to more sensitive data, a USB key such as Yubikey is recommended, as well as at least two factors of active authentication for users accessing the most sensitive data. For the Caterpillar customers social login through social networks such as Google and Facebook, due to the lower sensitivity of data they are accessing and the extremely easy user experience. These recommendations are not comprehensive, and as flexibility is paramount for the end user, other technologies are supported by all the companies that are recommended.

	Finger	Face	Social	USB Key
Employees	✓	✓		✓
Dealers & Suppliers	✓	✓		
Customers	✓	✓	✓	

Figure 4: Recommendations by Stakeholder Group

By having these technologies as a baseline approach, The Caterpillar Effect recommends requiring multi factor authentication for users who have access to yellow or red data. Multi factor could include a pairing of the above, one of the options with password, or one of the options with sophisticated passive authentication. Passive authentication can be very powerful and reduces friction. It can include timestamps, IP geolocation, keystroke patterns, etc. In addition, we suggest eventually offering SMS as a backup solution to eliminate any form of "what you know" authentication.

STATUS

Remaining Work

Caterpillar Effect will be sending Thank You's to Caterpillar, their faculty advisor, and the CMSC435 development team. In addition, the team will be following up to make sure that the client has received all relevant information and items (see 'handing over deliverables' below).

Presenting Results

On top of weekly calls with the client about the team's result progress, the team has made 3 presentations to Caterpillar employees this month (not including the in-class presentation).

1) The team did a dry run of the final presentation on December 1st with Rowland Nicholson and other Caterpillar senior personnel. The team received helpful feedback on improving overall narrative, which was used in order to prepare and improve for the following week.
2) Rowland was able to attend the QUEST Conference on December 7th and view the poster and presentation live.

3) On Monday, December 11th, Caterpillar Effect made the final presentation to Caterpillar management. The team was able to hear about their project was relevant to Caterpillar in the upcoming year.

Handing Over Deliverables

Based on the deliverables that Rowland Nicholson expects at the end of this project, the team plans to send a folder with the following categories:

- Findings and Analysis
    - Preliminary Online Research
    - Survey & Analysis
    - Prototype (Desktop and Mobile)
- Detailed Recommendations
    - Technology Feasibility Analysis Matrix
    - Company Checklist
    - Financials and Impact
- Final Presentation and Report
    - Present to Caterpillar Inc. managers and answer further questions
    - Deliver report with specific data and detailed recommendations

This folder will be shared with Caterpillar by next week (December 20th) over email.

Future Steps for Caterpillar

Caterpillar should begin integrating the Team's research by contacting companies for quotes and initial scoping with their technology stack. These quotes will inform Caterpillar Inc. of the price ranges the company offers and whether it fits within their budget. It will also detail to Caterpillar Inc.

After Caterpillar has received some positive response from the various vendors, demos with the technologies and Caterpillar Inc. Systems should be pursued. These will consist of testing the various technologies with some Caterpillar Inc. Employees.

Proof of Concepts are the next stage in the relationships between Caterpillar Inc. and a potential vendor. These are longer term demonstrations of the technology. Caterpillar Inc. has experience conducting Proof of Concepts, and did so in the Fall of 2017 with HYPR Inc.

An extremely important factor that Caterpillar must consider is the global and legal implications and risks. There are varying regulations on biometrics security and other information security around the world. In the European Union, there are regulations on the "right to be forgotten" and

various regulations about separating and cleansing user data. Additionally, some nation-states have more restrictive Internet policies than the United States.

- Asking for company quotes
- Demos with chosen companies
- Proof of Concepts
- Considering global and legal implications and risks

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APPENDIX

Appendix A: Feasibility Analysis Matrix

The Caterpillar Effect created a feasibility analysis matrix to rank all of the authentication technologies,by combining analysis of the three main categories of compatibility, user experience, and security. This information came from background research on the technologies, and weights were placed on different aspects, based on feedback from the team's client and their own evaluation of weights, to generate an overall ranking.

	Compatibility/ Installability	User Experience	Threat Coverage/Security	Overall Ranking
Password	100.00%	52.50%	34.17%	0.62222
Text/push notifications	77.14%	37.50%	60.00%	0.58214
Face recognition	44.29%	72.50%	81.67%	0.66151
Fingerprint	64.29%	76.25%	85.83%	0.75456
One-time password generator	68.57%	42.50%	73.33%	0.61468
Iris Scan	18.57%	71.25%	86.67%	0.58829
Heart Rate technology	57.14%	78.75%	90.83%	0.75575
SmartCard/RFID Wallet	40.00%	57.50%	79.17%	0.58889
USB Key	55.71%	67.50%	90.83%	0.71349
QR Code/Image Scanner	58.57%	38.75%	65.00%	0.54107
Keyboard Analytics	44.29%	71.25%	82.50%	0.66012
Virtual Token	68.57%	47.50%	80.00%	0.65357
Pattern	41.43%	52.50%	50.83%	0.48254
Voice	32.86%	65.00%	92.50%	0.63452
Social Passwords	100.00%	80.00%	34.17%	0.71389

Source: Secondary Research and Caterpillar Effect Spreadsheets

The color key for 'Overall Ranking' correlates to a green marking for values greater than or equal to 0.70, a yellow marking for values between 0.70 and 0.60, and a red marking for values less than or equal to 0.60.

Appendix B: Twitter Data Mining

The Caterpillar Effect used Twitter Data mining as a means to gather unfiltered public sentiment data on different authentication technologies.

Source: Twitter, date: 10/18/2017 - 11/06/2017

Appendix C: Scoring Matrix

Points	**Testbench Ease**	**Testbench UX**	**Testbench Speed**
1	USB Key	USB Key	Fingerprint
2	Face	Fingerprint	USB Key
3	Finger	Facial	Password
4	OTP	OTP	Face
5	Password	Password	OTP

Type	Sum	Out of	Score
Face	9	3	3
Iris	14	2	7
Voice	11	2	5.5
Smart Card	13	2	6.5
USB Key	10	3	3.333333333
QR Code	19	2	9.5
Fingerprint	11	3	3.666666667
Heart Rate	9	2	4.5
Pattern	23	2	11.5
Keyboard Analytcs	15	2	7.5
OTP	25	3	8.333333333
Password	26	3	8.666666667
Social	3	1	3

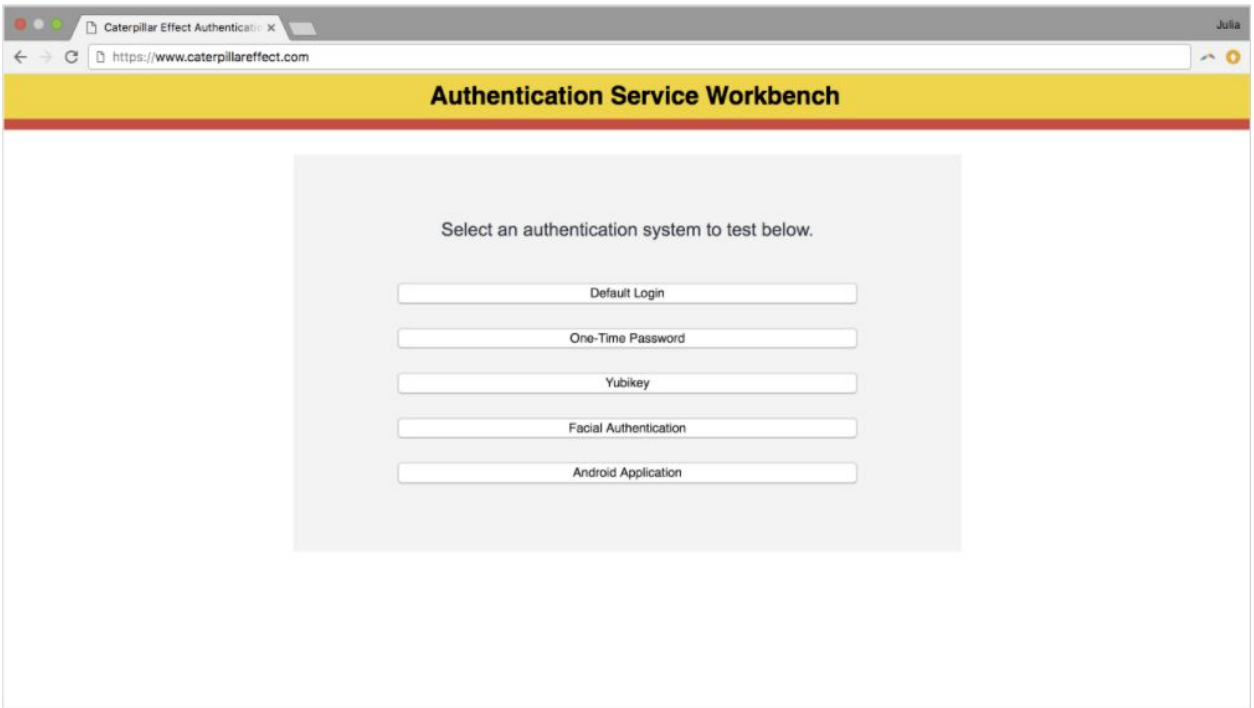
Type	Score
Face	3
Social	3
USB Key	3.333333333
Fingerprint	3.666666667
Heart Rate	4.5
Voice	5.5
Smart Card	6.5
Iris	7
Keyboard Analyt	7.5
OTP	8.333333333
Password	8.666666667
QR Code	9.5
Pattern	11.5

Appendix D: Authentication Prototype Results

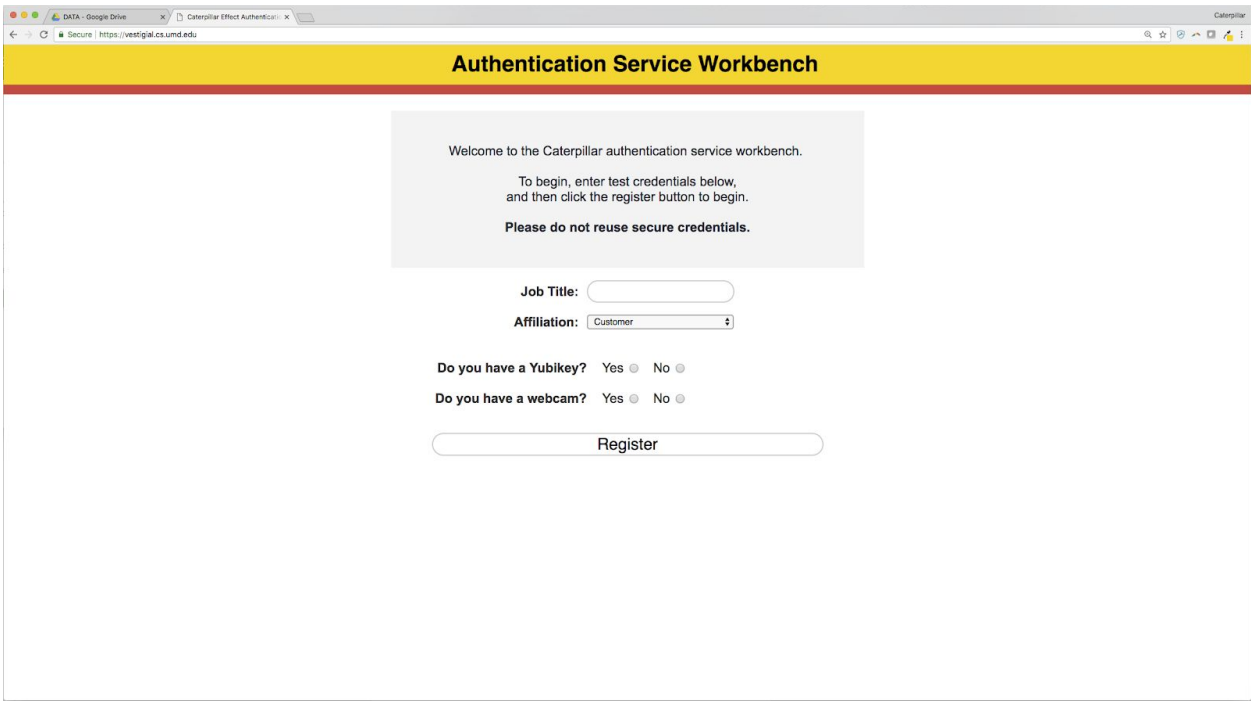
The Caterpillar Effect created a prototype alongside a computer science capstone team (CMSC435), in order to simulate user experience and other key metrics in a Caterpillar-like environment.

Source: Graphs generated in Excel by The Caterpillar Effect

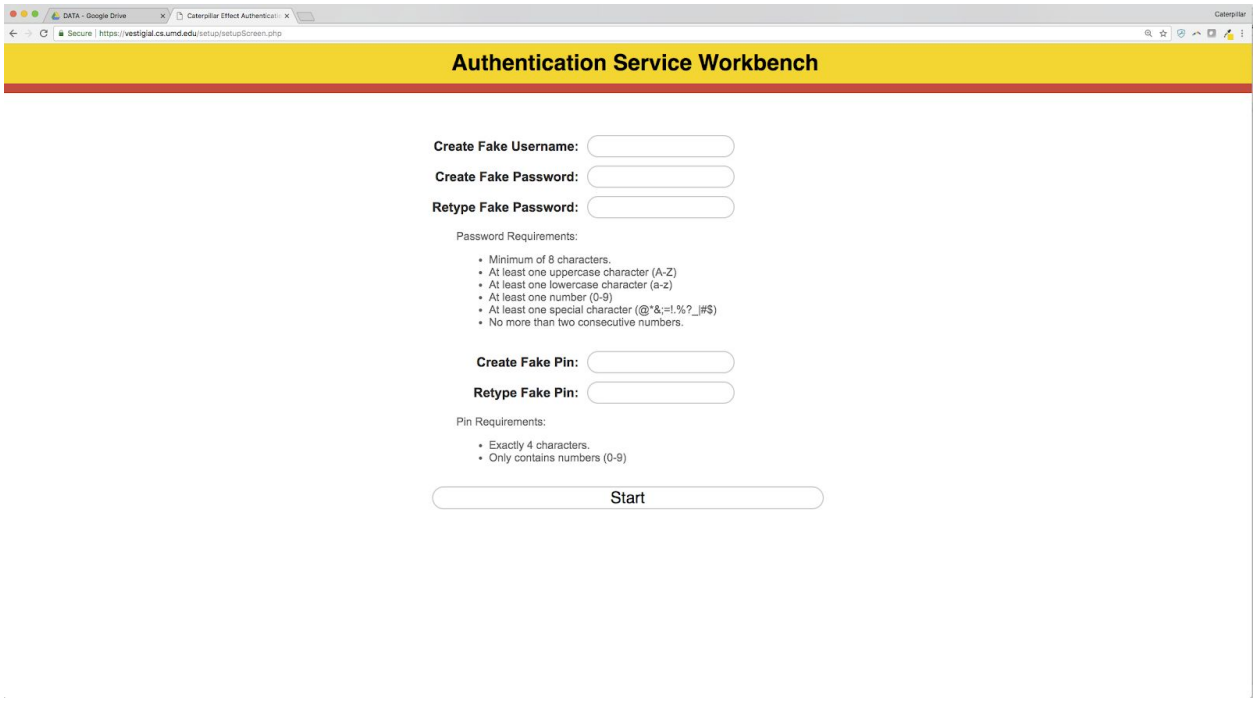
Appendix E: Authentication Prototype Screenshots



List of all login methods



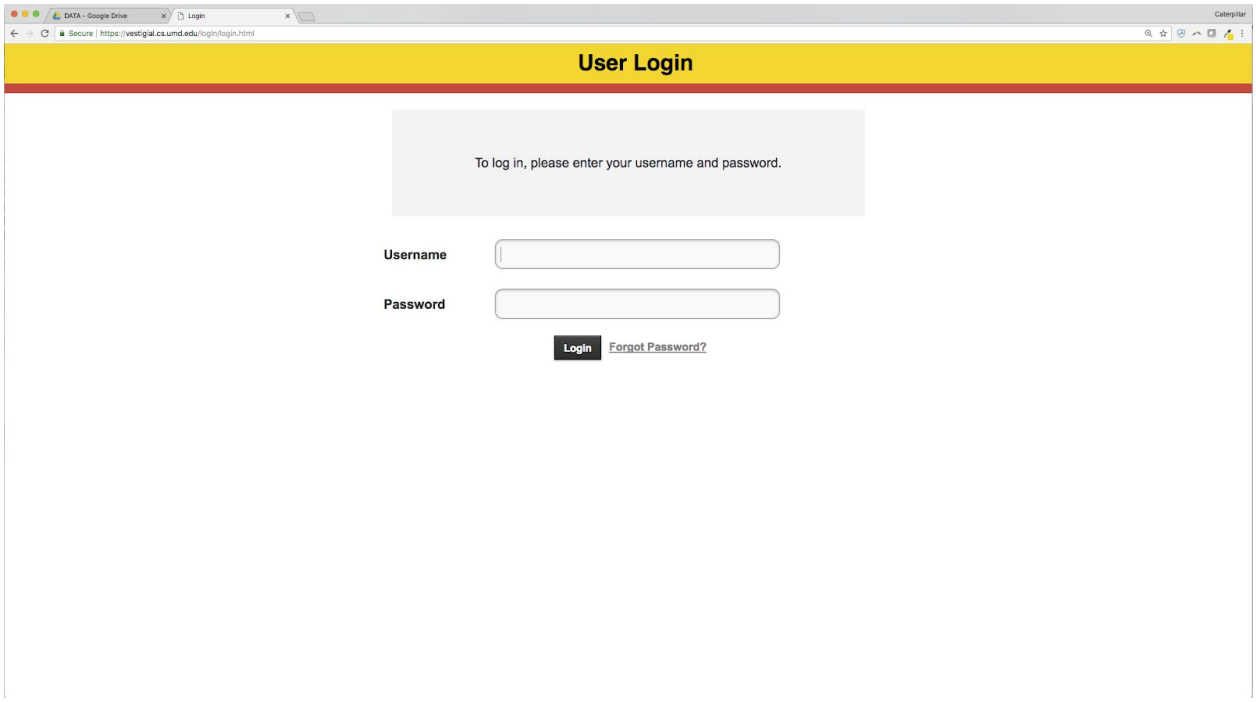
Initial Login Screen with demographic questions such as Job Title, Affiliation with Caterpillar, as well as what authentication hardware they have available.



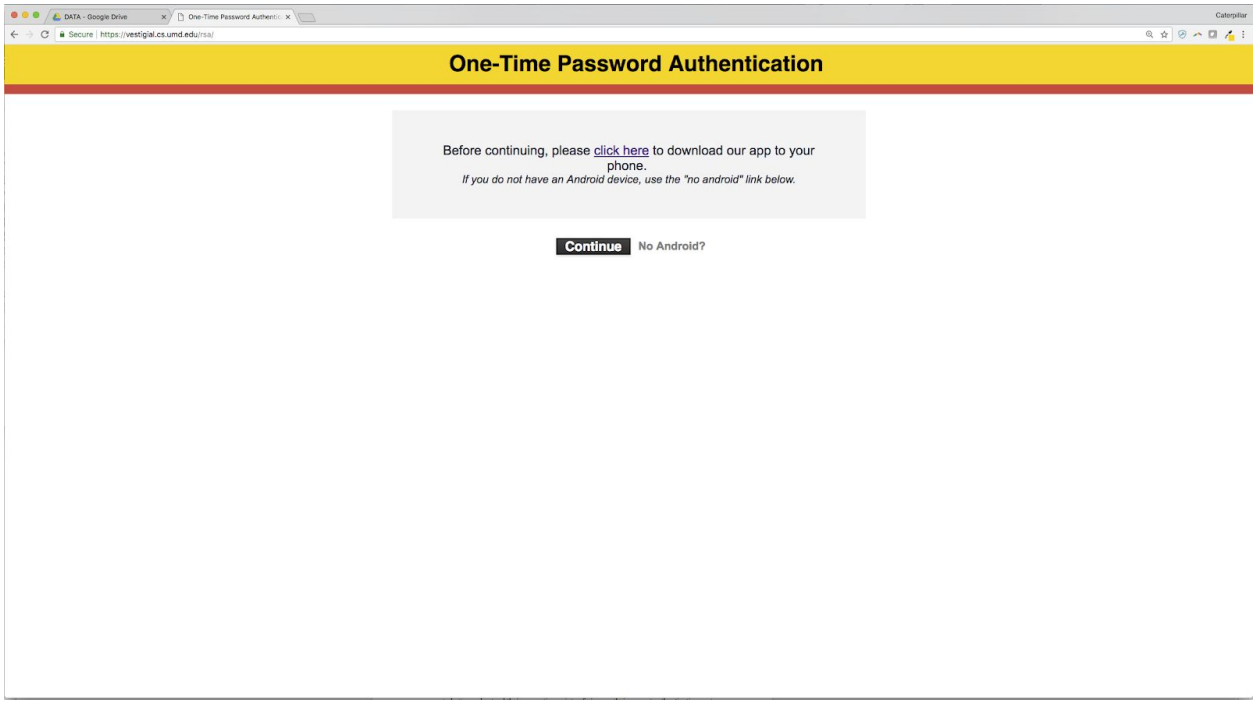
Registration Screen for first time users. This provides the necessary information needed to give the user a simulated authentication experience.

Demonstration of Yubikey Authentication Prompt. Once the Yubikey button is pressed, the text box fills with generated characters.

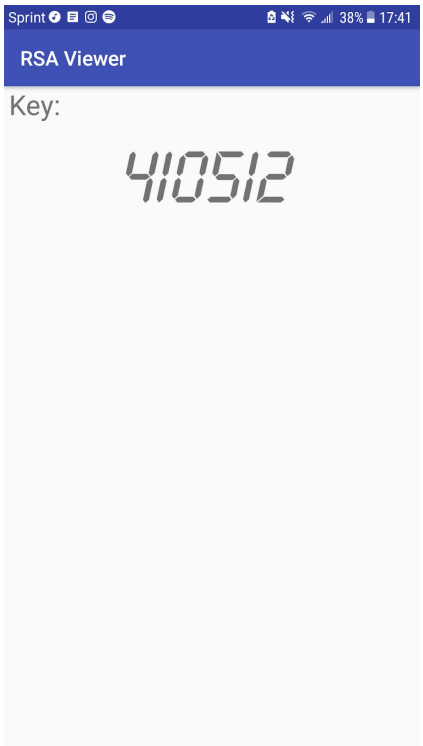
Demonstration of Yubikey Successful Authentication transition page.

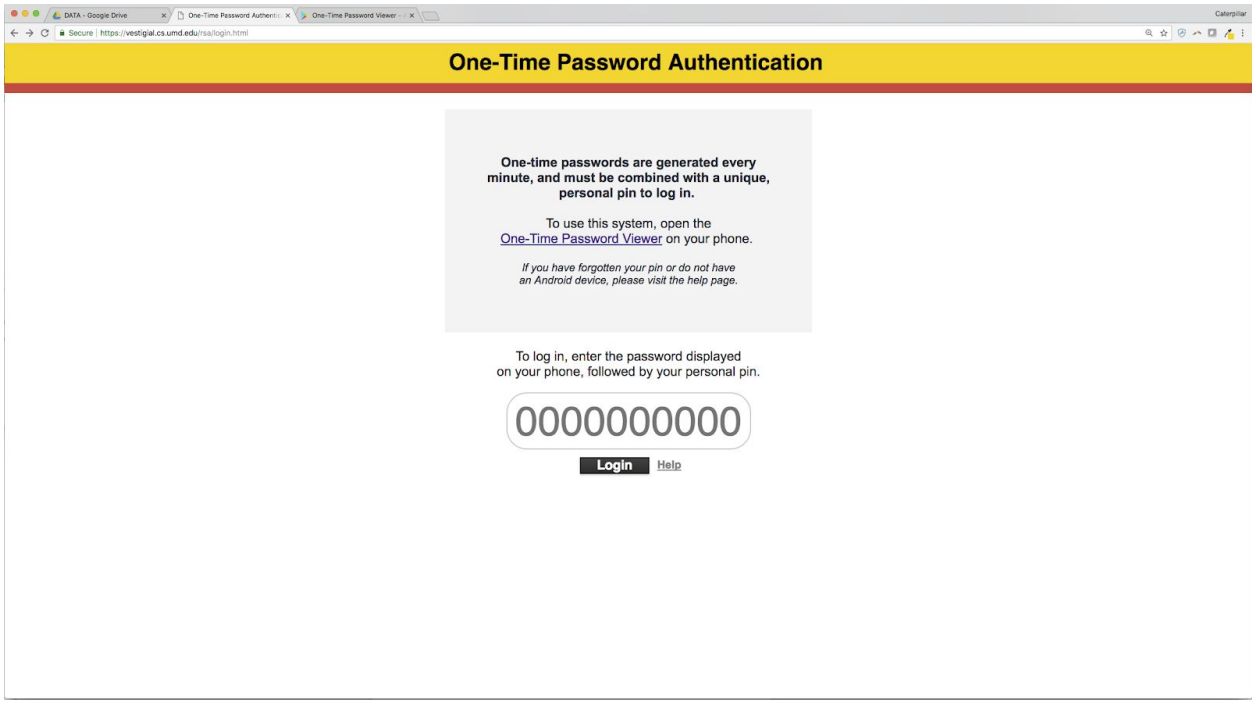


Demonstration of standard login and password authentication system.

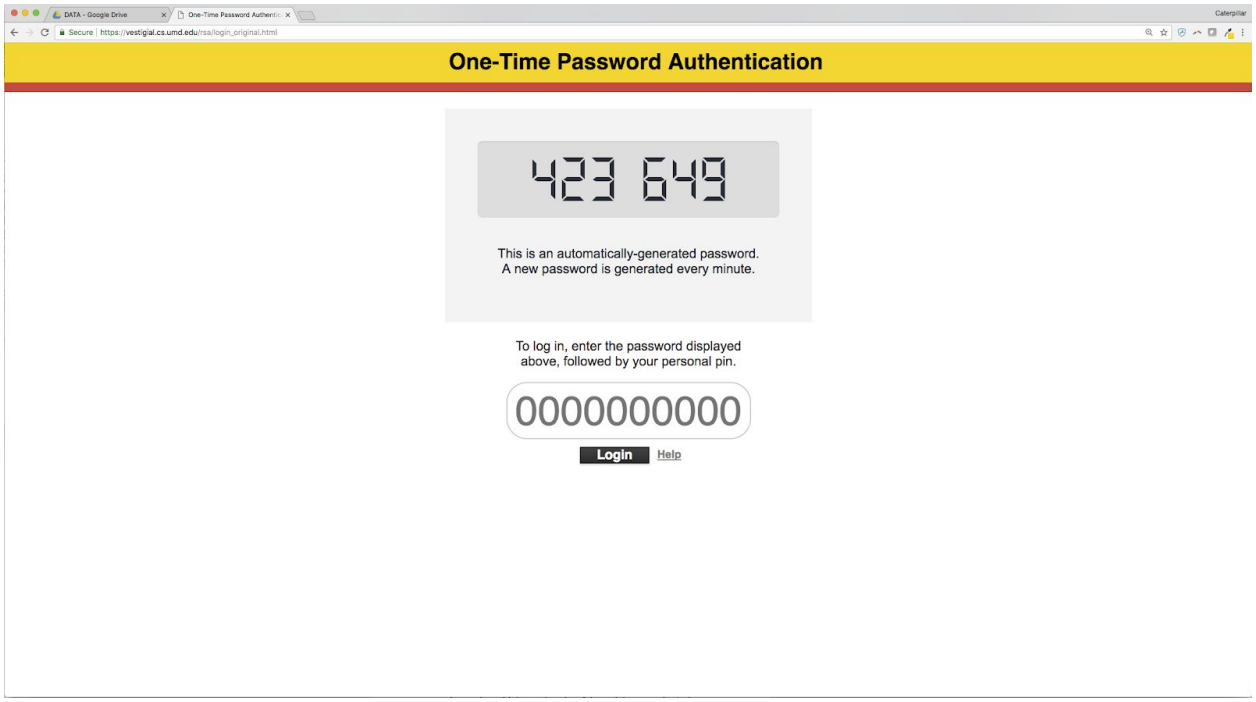


First screen of One Time Password (OTP) Authentication Experience. If users have an Android phone, then they may download an app, pictured below. If they do not an Android device, they may also authenticate over the web, as shown in the following image.





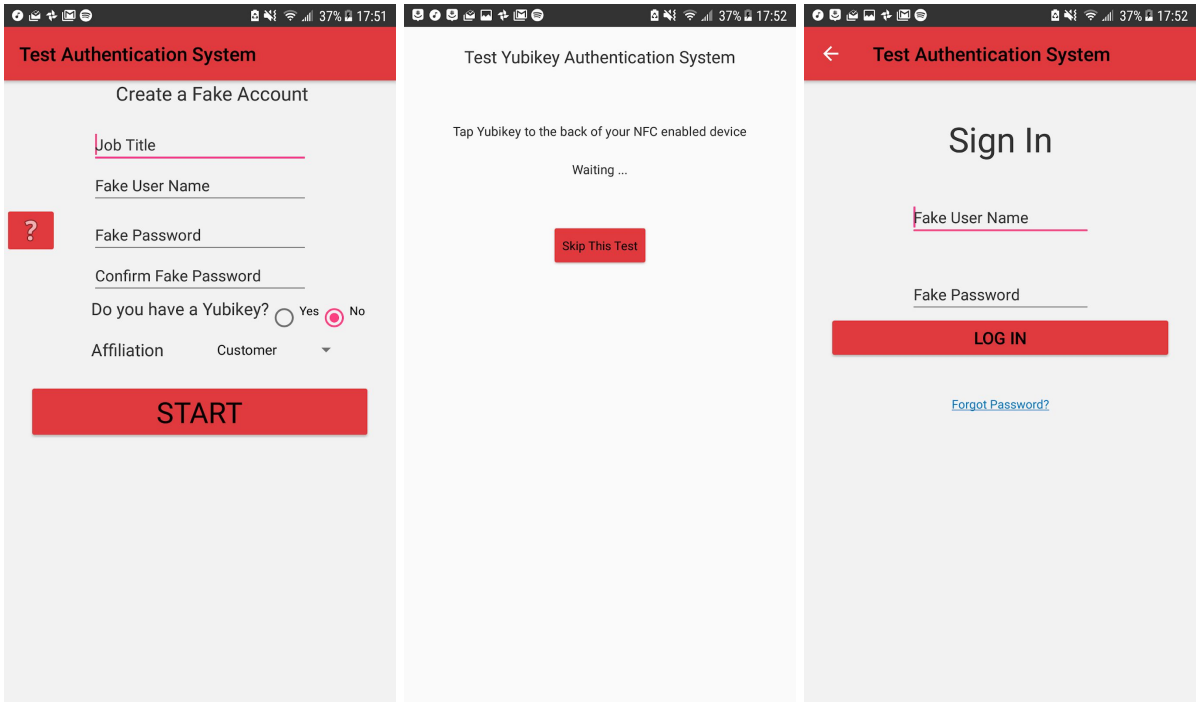
Demonstration of OTP Test with Android device. The first six digits are provided on the mobile device, and the final four are the users' pin.

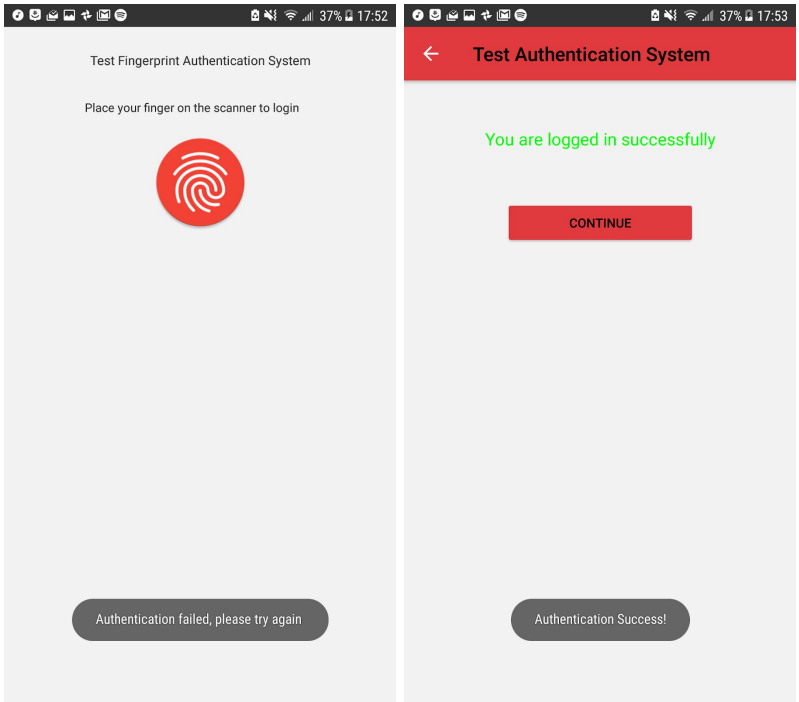


Demonstration of the web-only OTP Experience. It is obviously not as effective, however it did give some simulation of the experience.

Demonstration of Facial Authentication Experience. Facial Authentication would require users to first draw a box around their face on a setup screen, and then would be prompted for their image in the webcam before being allowed to continue.

Mobile Screenshots





Upper Row: Initial Login screen, Yubikey Authentication, and standard username-password
Lower Row: Fingerprint login, Login confirmation screen and survey

Source: All screenshots taken by The Caterpillar Effect

Appendix F- Key Caterpillar Stakeholder Survey Question Results

The Caterpillar Effect developed a survey and distributed it to different Caterpillar stakeholder groups to better understand their respective points of view on their current authentication system, new authentication technologies, and more.

Q- What is your tendency to adopt new technologies?

	Tendency to Adopt New Technologies					
Row Labels	Early Adopter - First 15%	Early Majority - First 50%	Innovator - First 2.5%	Laggard - Last 15%	Late Majority - First 85%	Grand Total
Dealer	34.38%	37.50%	14.06%	0.00%	14.06%	100.00%
Workforce	42.06%	37.30%	12.70%	1.59%	6.35%	100.00%
Grand Total	39.47%	37.37%	13.16%	1.05%	8.95%	100.00%

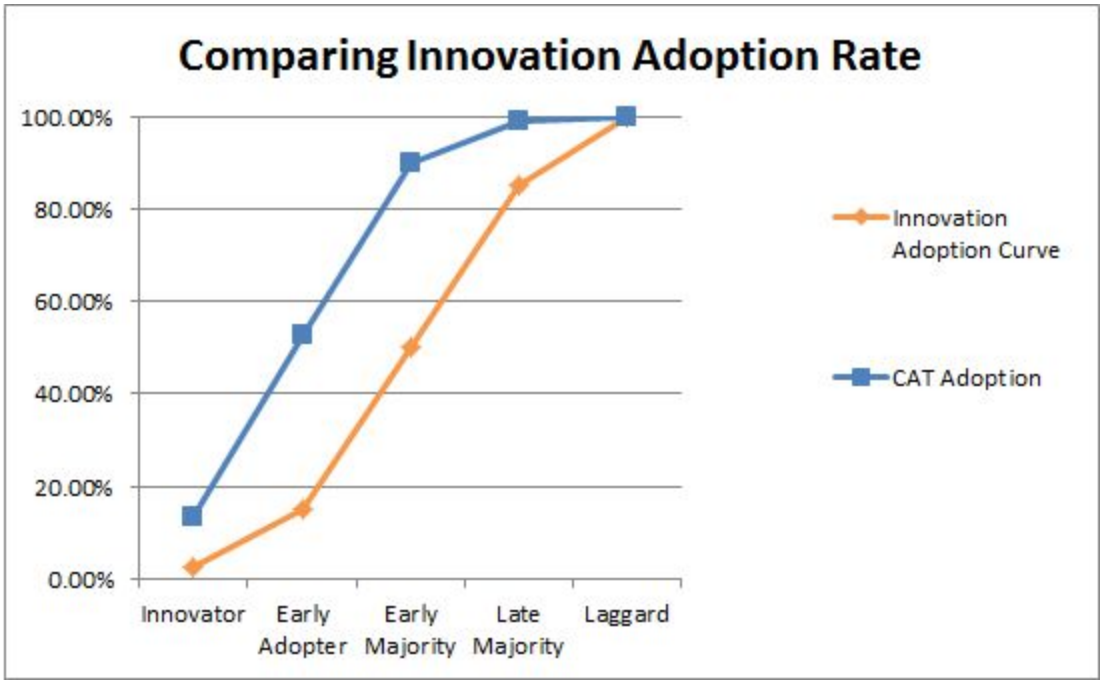


Figure 5. Caterpillar Innovation Adoption Curve (Source: The Caterpillar Effect)

(Note that the blue curve is based on the data we collected. The orange curve is graphing the defined cumulative percentages of innovator (2.5%), early adopter (15%), early majority (50%), late majority (85%), and laggard (100%), based on Rogers' diffusion of innovation theory.)

Q- How many times do you enter your password in a day?

	# Times entering your password per day					
Stakeholder	0-2	3-5	6-10	11-15	15+	Grand Total
Dealer	48.44%	34.38%	6.25%	3.13%	7.81%	100.00%
Workforce	1.57%	15.75%	22.05%	29.13%	31.50%	100.00%
Grand Total	17.28%	21.99%	16.75%	20.42%	23.56%	100.00%

Q- What is the highest level of data classification you have access to?

	Highest Data Classification					
Stakeholder	Non Confidential	Green	Yellow	Red	Not familiar with this classification	Grand Total
Dealer	2.60%	9.09%	54.55%	22.08%	11.69%	100.00%
Workforce	0.00%	4.96%	58.16%	33.33%	3.55%	100.00%
Grand Total	0.92%	6.42%	56.88%	29.36%	6.42%	100.00%
Count	2	14	125	66	14	221

Q- What is your level of satisfaction with your password experience?

	Level of Satisfaction					
Stakeholder	Extremely dissatisfied	Somewhat dissatisfied	Neither satisfied nor dissatisfied	Somewhat satisfied	Extremely Satisfied	Grand Total
Dealer	2.60%	11.69%	28.57%	44.16%	12.99%	100.00%
Workforce	10.64%	32.62%	24.11%	26.95%	5.67%	100.00%
Grand Total	7.80%	25.23%	25.69%	33.03%	8.26%	100.00%

Q- What is your level of satisfaction with password security?

	Level of Satisfaction					
Stakeholder	Extremely dissatisfied	Somewhat dissatisfied	Neither satisfied nor dissatisfied	Somewhat satisfied	Extremely Satisfied	Grand Total
Dealer	1.30%	12.99%	19.48%	44.16%	22.08%	100.00%
Workforce	3.55%	17.02%	26.95%	43.26%	9.22%	100.00%
Grand Total	2.75%	15.60%	24.31%	43.58%	13.76%	100.00%

Q - How long have been working with Caterpillar or using Caterpillar products / services?

#	Answer	%	Count
1	0-2 years	8.11%	18
2	2-5 years	13.51%	30
3	5-10 years	13.06%	29
4	10-20 years	33.33%	74
5	20-30 years	27.03%	60
6	30+ years	4.95%	11
	Total	100%	222

Q - Why are you not happy with CAT passwords?

The rules for a valid password have become so complex that it is difficult to manage. As an end result, I end up using practices that are also not recommended. For example, I leave the majority of my password the same and only change a few characters.

Too difficult to maintain

There are passwords everywhere, and the fact that we now require a password manager is an example of how this has become more and more complex every day.

The changing password process is poor.

Arbitrary rules for what is allowed and requiring to change them frequently means that everyone uses an easy password and just rotates the number at the end.

I have access to many systems and many different accounts. Each account requires a different password by policy with regular rotation and is hard to keep them all straight

In terms of functionality it serves its purpose, however is cumbersome to comply with password rules and expirations. Another important factor is the deal with second level access to other intranet apps, also multiple environments, separated identities and validations.

multiple user IDs and passwords for multiple systems/applications

Not easy to do, and depending on what systems you have access to, it can cause you to reset passwords a few times

As you have to change it regularly, you have to copy it somewhere which is not so good for security

users forgot passwords, users give passwords to co workers

takes time to input password. touch ID experience with iPhone is good.
Would like the ability to use passphrases. Passwords should not change every 90 days, you need to review the new standards they are changing. We have a password reset tool that we have purchased from the industry and CWS does not allow an interface with this tool therefore our employees have to reset their passwords independent to our tool. Not mobile friendly.
Too many of them. Need to avoid using same password for everything to minimize risk if breached, so literally dozens to remember.
Too simple to identify, need to have 2FA
There are still sites not included in CWS. The passwords are required to be more and more sophisticated...
Have to have a different password for almost everything and you have to change them every month or two so it is very difficult to remember them all.
Requirements keep changing. Multiple systems need separate passwords. Inconsistency with password requirements. Not always sure which sites I can "trust" with my password. Password expiration times vary between systems.
Having to come up with a new password every 90 days and trying to remember passwords is difficult.
Too many different passwords required. Difficult to remember all of them.
I have too many different passwords for different accounts and quite often get them muddled. I also find it annoying when passwords need to meet a certain criteria e.g. 1 upper case letter and must include numbers and symbols. Outside of work, I have to write passwords down for various different accounts
Changes too often.
Too many to remember and pw requirements for complexity.
Restrictions and changing, too many systems and too many passwords to remember
Multiple systems have different password requirements, with variation in update frequency. Seems like every 90 days I have to update to a new password (with different requirements), then spend a few hours figuring out the systems that the fix didn't propagate to. Then random 2 factor authentication. Then try to type that password on a mobile device (multiple times). The complexity leads most people to adopt less than secure practices.
Too many to keep track of
Needing to make them long enough to be secure and also type them in so many times a day acts as a tiny drag on everything that we do. Multiply a few seconds per password entry by dozens of times a day by tens of thousands of people, and you waste a lot of time.

Frequency of change requirements. Number of different passwords that have to be changed. The amount of time it takes to change multiple passwords for several systems, verification of changed passwords and frequent calls to IT when a password change fails is outrageous.

From a cyber protection point of view, each system we use would have should have a unique password. As I use far too many systems to remember a unique password for each system it is not possible to use an unique password for each.

Remembering multiple passwords that have to change every couple months is burdensome.

No sanctioned password manager. Poor rules that require difficult to create passwords yet often prohibit longer passwords. Screen savers that lock the screen during presentations.

Change too often. Requirement of not using last 90 passwords is difficult. Remembering a very long password is not the answer.

It literally takes hours to reset all my passwords every 3 months. Password syncing on Linux servers rarely works making you have to login to each server separately. Change Oracle passwords also requires you to connect to each instance one at a time. Some password length requirements are different so it can be difficult to remember them all.

Limits and constraints on ACF2 mainframe passwords are frustrating. Differing password length, security, and reset timing requirements across different systems and applications cause frustrating inconsistencies. Documentation and availability of password requirements and standards for systems to which I already have access is inconsistent and often difficult to find.

The current requirements for special characters/capital letters and constantly changing leads users to create rotational passwords. Terrible for security, but easy to remember.

Long and complex, not taking advantage of new ways/technologies to authenticate.

The complexity and the need to change so often. Getting to the point it becoming a pain to remember them, tempted to write some down.

Reason 1: They ask us to set, it requires multiple guesses which any one of it only the system accepts, the digits and rules to set is difficult to be followed. Reason 2: Requires remembering of the password after setting it is also quite tough, that too the very next day we reach office mostly wont remember what we set yesterday.

The changing of all the passwords so frequently has me at a loss, especially when I cannot synchronize them. I understand the reason, that's why I scored high on our password policy. It's just as a user, it is very difficult to remember all my passwords.

We aren't following a number of practices that seem to now be widely recognized in the industry: a) We require time-based password changes; b) We enforce composition rules that don't really improve security; c) Yet, we enforce a 12 character password length limit which does nothing but encourage people to use easily remembered/guessable patterns because of the composition limits and password length requirements; d) We have nothing in place to detect password compromise that might require a password change; e) Our training/education material still tells people to use leetspeak and other techniques that are no longer effective in today's world; f) We provide no inline recommendations for the strength of passwords in some systems; g) We provide no way for users to tell where they've logged in from (devices, browsers, operating systems, locations, etc.); h) All of this complexity and arbitrary rules to "improve password security", yet we have no tools such as a password manager that actually allow us to utilize secure passwords in an easy manner; if the security team wants people to practice better security then pick some tools that actually make it easier instead of only providing a bunch of directives without anything to enable those directives.

Constant need to change them and dream up something that meets all the requirements. Number of times I have to enter within my work day

There are too many to remember (CWS, ACF2, LAN, Teradata, MySQL, Linux, Oracle, HVV). Change frequency (90 days) is ok but it would be nice if it would be longer and be triggered by suspicious activity.

Due to the numerous systems, risk domain, and levels of access based on the activity, I have a list of passwords that I must maintain and a list of systems where password maintenance is local as well. So I have to manually keep track of where my passwords are and when they expire. I am also not very satisfied with the federation/single sign-on capabilities being used in our environment, there seems to be multiple federation services supporting various applications and I am constantly having to re-login to various systems.

I have too many different passwords to manage on a daily basis to access the different systems I need to perform my work.

We have to enter passwords all day long and should be using a tool such as LastPass.

passwords are fine.. Single sign on would be better.

Too many to track / remember; inconsistent requirements / length

Password complexity requirements are becoming more and more cumbersome, hard to remember, and are getting to the point that they have to be written down rather than remembered. It seems backwards to me.

I have to type them dozens of time per day. It gets monotonous.

Too many places and too many passwords to remember. Keep getting longer and more complex, making more likely to share between systems and therefore more prone to compromise. Also, in the personal world I need to share across my wife and I w/ same account - so that gets confusing.

Sad

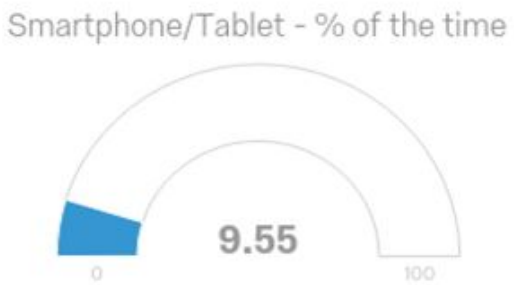
Too many of them to remember. Complexity rules make it difficult to create/remember them as well.

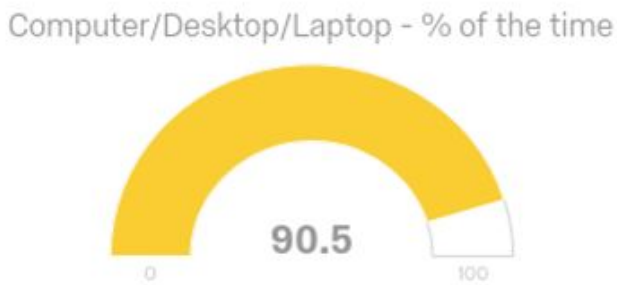
Changing and remembering them. Multiple passwords to maintain.
I have to type in my Password in various Systems at least 20 times a day. Why is single sign on not deployed. At least every second time Password sync between Windows and Lotus notes does not work. Mainframe Password does allow only limited Special characters
Annoying and not secure policies
I have to change my password too often. I use both CWS and ACF2 passwords, but ACF2 passwords have stricter rules on special characters. Passwords can't be too complex, because I want to use the same for both. They don't automatically sync (have to reset them individually).
Is not 100% safe
The number of different systems we have to access requiring security passwords with different change intervals and password structure formats, really slows the working day down continuously having to remember the password for each system and enter it every time you have to access it.
Password changes every three months and long passwords make memorizing them hard
We have single sign-on for some things, but there are just too many systems/platforms that require password entry and they all expire at different times. Can be difficult to keep all of them straight.
They have to be too long and changed too often. It is difficult to type correctly.
The requirement to change them on a regular basis does not comply with new NIST recommendations. No need to change unless breech or compromise is suspected.
I have to type them in to various systems frequently throughout the day. And have to change all of them every few months. Since I have multiple systems whose passwords are not in sync I just keep them the same to make my life easier and use the minimum length required. We do not have a way to store passwords for application accounts to be used by the support team so we store them in a password protected spreadsheet At home I use a password manager so all my passwords are long, complex, and unique.
too many different passwords to remember and reset. Too much time typing passwords repeatedly throughout the day
Too short. Too many complexities in required characters.
Too long to type dozens of time per day, Too difficult to remember passwords for over 10 systems
Has become cumbersome for our users
Becoming more difficult for our users

Q11 - From where are you logging into your CAT account? (Check all that apply)

#	Answer	%	Count
1	Customer Site	3.88%	15
2	Factory/ shop	3.88%	15
3	Office	48.06%	186
4	Home	31.01%	120
6	Public Wifi	10.59%	41
5	Other:	2.58%	10
	Total	100%	387

Q - What percent (out of 100) are you logging in to each type of device? Must add up to 100.





Q - How are you using your device once you are logged in? (Check all that apply)

#	Answer	%	Count
1	Email/Calendar/Collaboration	22.01%	171
9	Data Analysis/Management	14.41%	112
10	Engineering	6.05%	47
11	Financial	3.60%	28
12	Logistics	2.06%	16
13	Manufacturing	2.32%	18
14	Marketing	3.35%	26
15	Governance/Compliance/Legal	3.22%	25
16	Supervisor/Manager tools	5.28%	41
17	Project Management	10.04%	78
18	Services Support/Management	10.17%	79
19	Software Development	7.21%	56
20	Supporting Suppliers/Dealers/Customers	8.24%	64
8	Other:	2.06%	16
	Total	100%	777

Q - In general, what other types of login technology have you used?

Technology	Previously used and comfortable	Previously used and don't like	Interested in using	Never heard of
Passwords	159	32	14	1
Text / Push Notifications	138	25	17	11
Face Recognition	9	34	100	12
Fingerprint	111	25	55	4
Iris Scan	6	16	101	25
Heart Rate Technology	6	9	50	85
SmartCard / RFID Wallet	58	25	66	19
USB Key (e.g. Yubikey)	23	34	65	37
QR Code/Image Scanner	46	28	58	28
Keyboard Analytics	4	14	40	90
Virtual Token (e.g. one- time password generator on phone)	101	32	31	17
Hardware Token (e.g. RSA)	92	43	17	16
Voice Recognition	16	24	90	20
Pattern Login	44	26	46	50
Other:	3	1	4	3

Other:

Text
Multi-factor authentication
Two Steps verification
Social Password

Q - How do you think Caterpillar can improve its authentication/login system?

I think Caterpillar should make a serious financial investment in alternative solutions.

I would love a fingerprint scan on my laptop. I have so many passwords to recall that I don't keep one on my phone or my laptop at home. I use many programs within CAT and everyone has to have a password and login. With that a change is needed every 30 days. I have too much to try to remember, my bank account number, my ATM card password, my call in to my bank password, my password to get online to my many accounts. I wish CAT would install a fingerprint password for the laptop and then for the programs, no log in just a 4 digit password minimum. It's too much to remember passwords to each program and to recall the special character used each time.

Please use any secure system other than current strong password system

Move to a cloud based managed service provider to provide directory support

I think that Caterpillar has good login system. It is very comfortable and intuitive.

Think of impact to daily change required for users before selecting a system to be implemented. Once it is purchased, we will be saddles with any downside because of necessity to maintain cost structure.

I think that the present conditions are enough.

By educating users and having a strict access policy

Technologies and keep changing. Data security is vital. Definitely Caterpillar can improve authentication and login system.

Yes, it would be good to start using alternative technology that is available to us.

Allow federated login to enable dealers to also leverage Caterpillar's security model

Add one more step after password

Use of captcha authentication System

By being easier and faster without compromising security

By Federation/SSO with Dealer-Systeme (e.g. Azure-AD etc.) - if another solution is chosen then 1. OTP (SMS or AUTHY) 2. Fingerprint for Mobile 3. For Laptops etc --> certificate based

A single authentication method for accessing all resources a user may be subscribed to.

An easy to use and not very complicated to change and remember.

From an administrator's perspective: There are different systems to maintain accounts and licences. Our centralized IT department working for several dealers is struggling to create and maintain accounts. The security of data is an important subject, so we want to make it easy to use, but the data must be well protected. From a user perspective: More use of one userID and password, and any technology where the password does not have to be memorized is welcome.

consolidate userid/passwords
2 factor
For mobile can use fingerprint tec and for laptops virtual token
The password system his working fine in my opinion
I use the dealer portal and have a login and password which is very standard. I do not use the portal enough that it important to me to have a different method.
keep it simple and have it aligned throughout the organization
I think that improvement is not needed for the moment
Adding a second level of authentication.
They keep the security configuration on their servers. By them generating the temporary passwords, and emailing them to the users, there is not much to improve. Token generators security, is useful, but very hard to keep track of the generators and ensuring that terminated users are having the tools removed from there access immediately.
Reduce the number of accounts and Systems to create access
Force stronger password. Not allowing old password / too similar reduce time before laptop locked out. Force to have specials caracter
Biometric Solutions
Yes - two factor authentication
SSO with Dealer to not enter any password anymore.
It would be easier to use fingerprint or similar. Would be interesting to provide a way for dealer to use this authentication/login system for our own customer portals
Make it easier for users to reset password. Passwords are not easy to remember, you can ask IE to remember but then if you don't type it often, you will forget it.
CAT system's better to use factor authentication system....
OTP with text messages or app generated pwd
Have one password for the different applications like PCC, SIS, Biller direct etc.
2 factor authentication system or biometrics support, such as fingerprint can be introduced.
Multi biometric systems are a good complement to traditional passwords or simple fingerprint systems
finger print with backup password

The Current Single sign on technology is quite robust. You can introduce One Time Password Phone app integration in conjunction with typical password. That will not only boost security but user awareness of password importance as well.
By using biometric authentication on smart devices.
I like the app how the token temporally en smartphone and iPhone
Using body (finger, face etc) should improve.
Technology is every improve day to day. Most of system have easy to use and more safe for a person. So, CAT can improve more convenience for end user.
Currently, Authentication system includes lower/upper case characters, numbers and special characters such as (*, #) that I'm aware of. I believe the system can be improved if instead of allowing (upper case OR lower case) characters, the program can require (for instance) lower case characters AND AT LEAST one upper case Character (or viceversa, upper case characters AND AT LEAST one lower case character) That fact will add some extra protection to the password itself
I don't think that Caterpillar needs to do anything more than what they have. We are a dealer and have limited access. This question might be better answered by Cat employees. If you were to add a layer to a dealership, two factor may be the best choice.
With a high level of security and different ways to get authenticated.
Two factor authentication using smart phone app or SMS text code delivery
current is okay
Adding second factor (e.g. soft token).
Fix random bug where system loops asking to change password.
Don't allow people to save their password in their computer, so they won't forget it. Usually when they will forget it once they finish to type it and save it the first time. Save password make system weak also because anybody can access to any caterpillar resource if they have access to the windows profile of the user.
One login to get to all CAT applications that I'm authorized to use
Add 2 FA for secure sites
Caterpillar is lengthening passwords to 12 characters in the near future. The entropy based on overall password length vs. the available character set vs. forced use of particular numbers of characters from particular character sets can be fairly well estimated. The ever-increasing lengths of passwords should be balanced by forcing entropy in other ways.
Fingerprinting or voice recognition would be faster than typing passwords.

Learn from other companies...

Having a system that would not create the need for a password change every 3 months would be ideal. Be it a push notification or fingerprint, it would make life much easier from an everyday work experience perspective.

There is new technology out there, as evidenced by the options on the previous question.

The current authentication/login system seems to be effective. The most cost effective system needs to be considered.

Coordinate ALL the systems to use a single sign-on, instead of just the majority of them

Make it simpler.

Use common process across all applications.

Since I'm not a security expert I don't really know. It seems like there should be better ways to verify who I am than passwords, which is just something I know and can be shared. Something like a fingerprint scanner is hardware and costs extra, maybe there is a technology to use the existing camera in laptops, pads and phones to scan a face, iris, fingerprint, etc.

Not sure. I do like using the VIP Access.

It can definitely get with the times and use something like an card chip or RSA token to stay logged into things. A lot of caterpillar software and is cripplingly out of date and hinders job performance, and the things they do update, they keep the older one that people keep using and you get crossed between the two.

Fingerprint services could be worth using. Or maybe some kind of wireless fob that unlocks devices when its in close proximity. The only downside I see with using these technologies, is that when log in attempts fail they often request using a backup password. The problem with that is, the individual has become so used to using alternative methods of logging in, that they can no longer remember what their password is. I have done this many times at home with various accounts, and I have to recover by receiving a temporary password via email.

Require longer passwords, allow arbitrarily long passwords, remove special key requirements, remove password expiration.

Develop trusts between CWS and ACF2

While I understand the need for passwords, I think most of them should be interchangeable with systems. The less logging in, the better.

Use multi-factor not just when on a different network.

Not sure. I think our current system could use improvement but security is incredibly important

One very secure login , with everything tied to that 1 authentication.

True, universal single sign on

A non-password way to authenticate would be nice, especially if it didn't involve carrying around another physical item (phone, card, usb, etc.). Biometrics? I like the fingerprint reader on my phone.

By requiring more complex passwords, which is being rolled out shortly anyway.

Fingerprint or something that is quick, but still as or even more secure. We are being asked to have even longer passwords, so I will become even more interested in alternatives!

Increase the time required in between password changes and work towards a true single sign on system.

There are a lot of people who still leave their desk tops unlocked when they walk away from their cubes. A seat sensor to lock PC if you get up for more than a few seconds might be a good idea.

Any system that reduces the amount of time one's password has to be entered per day.

More biometric authentication like fingerprints. Most of us do not use shared computers so once I'm logged in to my computer, I shouldn't have to login to each system - SAP, Teamcenter, etc.

Allow greater access on personally owned devices. In particular, I want to be able to remote into my Cat owned computers from home such that I don't have to haul home an 8# laptop every night on the off chance I might need it.

Different passwords for LAN and CWS coupled with the length and update frequency are causing bad behavior of writing these down. There has to be a different way to authenticate someone. I like the use of VIP to confirm i.d. remotely.

Get rid of the special character requirements that make passwords hard to remember and instead enforce long passwords.

Develop/implement more robust pwd and authentication options. Consistent communication and training on implemented solutions for impacted users.

Adopt single login instead of multiple logins

Instead of just forcing us to make passwords longer, lets use technology to make login more secure and easier at the same time.

A form of biometric finger scanning would be useful to easily confirm user identity.

One way/password to log into all systems and reduce/eliminate need to log in throughout the day.

I don't mind using 2 factor authentication with VIP. Not sure if that is a good solution in place of passwords altogether or just the initial entry.

I think the 2 factor authentication is a good tool when working away from the office. I do have to enter my password a lot in a day, mostly to unlock my computer after I've stepped away from my desk, but I don't know what would be a better solution.

2 factor (push notification or finger print and/or password) everywhere with shorter simpler passwords

Stronger passwords like passphrases and two factor authentication

I'm happy with the set up as it is today.

I think it's pretty good to the point of being a necessary pain (hate passwords and needing to change them every 90 days), so wouldn't recommend any change. I do not like carrying a separate device (like USB key) for authentication - at the moment we use a symantec app on the smartphone for remote log-in, text messages with security code for verification etc - which is as far I'd like to go with regards a secondary device. If this could be integrated fully for all kinds of login system - then I'd just prefer to use a facial recognition/ fingerprint sensor from smartphone - so that typing passwords are not necessary.

I think Caterpillar could user passphrases more and maybe different steps/Systems for authentication. However, I would feel rather uncomfortable with fingerprints, face recognition etc. Additionally there would be legal barriers to this in regions outside the U.S., especially Europe, so country differences need to be considered.

Make login simpler, so that we are not tempted to keep saying yes when ask if password should be saved. I sometimes do to save time.

Pattern might be better, or new technology is worth a try. Easiest way is giving a preset password.

I think that they are doing a good job with it as it is now.

Two-factor authentication is a step in the right direction

Provide technologies that work WITH our security directives that make it easier for people to manage their authentication sessions in a way without worrying about passwords (e.g. password managers, fingerprint or other biometric-based authentication, etc.). Provide the ability to see the devices that are logged in and be able to logoff sessions that are no longer active from other devices. Provide the ability to be authenticated/authorized for longer sessions. The more often users have to enter a password, the easier they will make their passwords to remember using password anti-patterns. The security team needs to recognize that adding more and restrictive controls and directives don't improve anything; people's nature will be to work around more restrictive policies regardless of how you communicate it. Utilize new technology capabilities to make it easier for people to comply with the directives instead of looking for more ways to make more restrictive policies. We're just covering up the problem with a false sense of security because we think the policies are "more secure" but it doesn't matter if they're so difficult to work with that people find a way to work around them. Make the directives and policies secure enough, but provide the tools and capabilities along with those policies that actually make it easy for people to comply. Just adding policy and directives without enabling people to work better, won't change behavior or culture. Training and education to address ignorance isn't a bad thing, but pick some tools and technologies that helps make progress towards making ignorance irrelevant in order to address all aspects of people, process, and technology.

Embrace and distribute new technologies

Reduce the number of times I need to enter the password from within work

Use methods other than passwords for some authentication.
Improved company wide single sign-on, possible federated with third-party provider, include option for biometric and/or multifactor smartphone token or text back.
By offering a better SSO solution, as of today we have to authenticate multiple times with CWS Id because we're using different services without being able to pass the security token from one website to another (probably less secure in certain ways, but definitely more convenient from an pure usage perspective).
To extent passwords are required, have one universal password.
Use LastPass. Not only does it manage your passwords, but it enables you to have much more complex passwords that are all different from another and you don't even need to know them. I have over 400 person passwords in my LastPass. It works great.
Investigation into the facial recognition and Heart Rate technology sound interesting.
Single sign on.
Assure that login doesn't depend on battery power in a cell phone or other device to receive text or provide a code.
Simplification while integrating newer / proven technologies
It can find a way to make it more secure, but less cumbersome to use. No one wants to do something that makes Cat vulnerable, but we also don't want to waste time just trying to get access to systems/apps needed to do our work.
Fingerprint
Enforcing some criteria that can't be written down and shared with others
Device should just "know who you are" and let you in.
Allow for 1 authentication and then use it across all systems
Needs to be secure but easier to use than generating/remembering passwords.
Look to implement newer technologies / methodologies for data access.
Move away from passwords, wastes too much time
Use USB-connected fingerprint and/or iris scan authentication technology.
Something unique that does not have to be changed all the time.
Very happy that Cat investing to extend single sign-on to additional apps.
Use multiple layers of security

Simplified end user experience
For most - if not all - Confidential Red information, 2FA should be a mandatory requirement. Other than that, I'm not really sure.
Need to find a simpler way to login that still incorporates two-factor authentication.
Educate Software requesters about the risks of using Caterpillar data/policies in Third Party Software. We have varying levels of confidential data in External sources which require additional logins outside of CWS. These requirements are often not as stringent and thus less secure. A Third Party vendor is not as likely to take Caterpillar confidentially as seriously as an Internal Software team which exposes our data to additional risk.
They are in the process of making passwords more secure.
Yes, there are issues with the functionality that need to be addressed. Technically it would be nice to have more consistency in the implementation and integration of capabilities like SSO.
I'm not sure. But, I am open to new technologies, I just don't have much knowledge of them or how easy / difficult they would be to use.
Seems the passwords are getting longer and longer. Moving to a pin with 2-factor, or something biometric would make that less painful.
I think something that would be biometric and common across all products. Also, need to consider how this would work with things that are second hand authentication such as firewall authentication.
Improve Single sign on, so that I do not have to enter my Password so often, especially as we are no changing to 12 character long Passwords.
There is so much confusion right now with CWS, LAN, different QA passwords. There should be single sign-on established. The wifi is a common one, it should be ID specific.
I like the two factor authentication at home using VIP. At work, I think that a fingerprint scanner connected to my keyboard would be perfect.
Enable modern authentication methods such as biometrics - fingerprint, retinal scan, facial recognition.
Maybe use face recognition instead of password to login to your machine
Improve and implement bio passwords
Make it simpler
Two factor authentication is the first thing that comes to mind. I think it would need to be something that is very easy to use a lot of times per day since users typically sign in to their computer or various applications easily 25+ times per day. Personally I think a solution like an RFID card or YubiKey which users would use in conjunction with logging in would be the best kind of improvement in authentication.

I believe they are doing a fine job right now. As long as they have an experienced, security-savvy group staying on top of the latest methods, then we are in good hands.

Fingerprint scanners are an easy, fast and reliable way of authenticating.

Using fingerprint devices on laptops instead passwords or password manager like LastPass.

One standard change interval, one standard format in terms of letters, numbers, symbols and length. Alternatively use a two factor approach in terms of the RFid Id cards and a password to SSO to all general applications and only require an additional password entry on Red/Yellow security classification information systems.

Caterpillar is like any other - using passwords as the main security measure. This needs to change on a large scale to protect companies and personal data.

Working good the way it is, stay with passwords, not interested in using my iris to login

Find a solution that authenticates and provides single sign-on capabilities to corporate systems. I do like the idea of using a technology that is unique that wouldn't require changing every 90 days like current passwords do.

Do not ask us to login multiple times like it happens sometimes for Intranet or office 365 apps

Add more factors when possible.

Using Password Managers could help for teams to store application passwords. And Password Managers allow employees to only need to know one or two passwords so they can be longer and more complex. Having to change passwords frequently is no longer recommended by the National Institute of Standards and Technology and has been shown to not help with security so we should stop requiring password changes every X number of days.

I generally think it's fine as is.

Establish consistent authentication requirements across systems. Just as in our private lives, trying to manage multiple authentication requirements between systems results in unsafe tracking of passwords due to fear of getting locked out of accounts.

For the most part I think that passwords provide good security as something that I know, add in 2 factor authentication and it seems like a good solution. The current "hassle" that I have with passwords is entering it each time I return to my desk to unlock my system. Entering it each time I visit certain sites that are a cloud service (box, taleo sites for career related) and keeping track of the password for 10 different systems/applications that I use. It would be great to go SSO with 2FA but some legacy systems (Mainframe accounting system) may not be capable of that.

Consolidation of accounts across systems

Require multi-factor authentication. Reduce password complexity. Increase password length or replace with biometric authentication.

Increase the number of resources protected by MFA.
Modernizing. Integrating and federating logins across systems
Satisfied with current system
Explore other options
Give users more choices

Source: Qualtrics Survey Results, responses from assorted Caterpillar stakeholders

Appendix G: Company Rankings and Recommendations

The Caterpillar Effect developed a recommendation list of companies that Caterpillar should consider partnering with to provide for their authentication needs through evaluating their technologies offered as well as additional requirements. The team also developed a watchlist of companies to keep an eye out for in the future.

Table 4: Company Ranking Matrix

	SecureAuth	Okta	SecureAuth	Deepnet Security	SafeNet (Gemalto)	Bitium	SaasPass	Nymi	Safran Identity & Security	Nuance
Password	x	x	x	x	x	x	x		x	
Pin		x				x				
Text Push	x	x	x	x		x	x		x	
Face Recognition		x		x		x		x	x	x
Fingerprint/Touch		x		x		x	x	x	x	
OTP	x	x	x		x	x	x			
Iris Scan									x	
Heart Rate								x		
Smart Card				x	x				x	
USB Key	x	x	x	x	x	x				
QR Code										
Behavior Analytics									x	
Virtual Token		x			x					
Pattern					x					
Voice		x		x				x		x
Social Passwords	x	x	x							
Bluetooth					x		x			
Requirements										
SSO	x	x	x	x	x	x	x		x	
FIDO	x	x	x	x	x	x			x	
Cloud	x	x	x	x	x	x			x	x
Analytics Reporting	x	x	x	x	x	x				x
Cordova		x link					x			
MFA (incl. passive)	x	x	x	x	x	x			x	x
Scalable	x	x	x	x	x				?	
Cost	5$ per user per month	$2-3	5$ per user per month	32$ per person/yr when 1000 users	1$ per month for 2500	8$ per user per month	$40-60 per YEAR			

Source: Assorted secondary research from company websites- See References section